2-step PCR mixture and conditions (Barcoded-head primers for seqs pooling)

Huang MEGA Lab (NCKU)
v1.1, 2026-10-21; environmental samples (soil, water) and mixed templates; hand-maintained since 2026-10-21

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Use when: amplifying from **environmental samples** (soil, water) and other **mixed templates**: the 1st PCR (gene-region primers) gives a product to check on gel before the 2nd PCR adds the barcodes, so inhibitor-rich or low-copy extracts can be rescued. For single strains (one or several loci) use the one-pot PCR instead.

Abstract

PCR mixture and condition (2X SUPERGREEN PCR MASTER MIX)

PROCEDURE

1. Wear glove, clean up the working bench w. 1% bleach

FOR 1′ PCR HEAD-PRIMERS

2. Prepare 1′ PCR master mixutre for **head-primers** (**prepare 1.2X of solutions for pipetting error if needed**)

PCR mixture for head-primers for each reaction

Component	Volume	Volume (1.2X)	Final conc.
PowerPol 2X PCR Master Mix	**12.5 µL**	**15 µL**	1X
Forward Primer (10 µM)	**0.5 µL**	**1.2 µL**	**0.2 µM**
Reverse Primer (10 µM)	**0.5 µL**	**1.2 µL**	**0.2 µM**
ddH2O	**10.5 µL**	**11.1 µL**	-
Total volume	**24 µL**	**28.5 µL**	

Component	Volume	Volume (1.2X)	Final conc.
2X KAPA HiFi HotStart ReadyMix	**12.5 µL**	**15 µL**	1X
Forward Primer (10 µM)	**0.75 µL**	**0.9 µL**	**0.3 µM**
Forward Primer (10 µM)	**0.75 µL**	**0.9 µL**	**0.3 µM**
ddH2O	**10 µL**	**12 µL**	-
Total volume	**24 µL**	**28.8 µL**	-

Note Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

3. Mix the 1′ PCR master mixture gently by pippeting. Quick spin the tube.

4. Transfer **24 µL** 1′ PCR master mixutre in 8-strip PCR tubes.

5. Add **1 µL** DNA template in 8-strip PCR tubes, resulting in a **25 µL** reaction mixture for 1′ PCR.

Note **Negative control** contains only **24 µL** master mixture but not DNA template

6. Mix the reaction mixture gently by tapping the tubes. Quick spin the tubes.

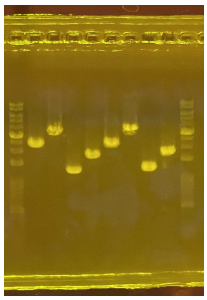
7. Carry out PCR using the following condition:

1′ PCR condition for **head-primers**

Step	Temp	Sec	Cycle
Initial denaturation	**95 °C**	180	
Denaturation	**95 °C**	30	**25 cycles**
Annealing	**60-66 °C varied (b)**	30	
Extension	**72 °C**	180	
Final extension	**72 °C**	420	
Preservation	Preservation	**4 °C**	∞

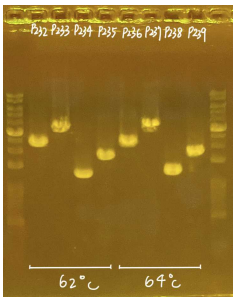
b. Annealing varied, 60-66C is working; Refer to 1′ PCR primers for annealing temperature c. 1kb ~ 1min extension; enough time allow full extension of sequence

8. Carry out **electrophoresis** for inspection of DNA products



Gel before markdown

9. Markdown wells and upload the pictures to the Lab Google drive



Marked gel picture go to the Lab Google drive

FOR 2′ PCR BARCODED-HEAD PRIMERS

10. Prepare 2′ PCR master mixutre for **barcoded-primers** (**prepare 1.2X of solutions for pipetting error if needed**)

PCR mixture for barcoded-primers for each reaction **(NO PRIMERs at this point!!)**

Component	Volume	Volume (1.2X)	Final conc.
ZEJU PCR Master Mix	**7.5 µL**	**9 µL**	-
ddH20	**5.55 µL**	**6.66 µL**	-
Total volume	**13.05 µL**	**15.66 µL**	-

Note Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

11. Mix the 2′ PCR master mixture gently by pippeting. Quick spin the tube.

12. Transfer **13.05 µL** of the 2′ PCR master mixture to 8-strip PCR tubes.

13. Add **1.2 µL pre-mixed barcoded-head primers** (Forward + Reverse) to each PCR tubes.

14. Add **0.75 µL** of **1′ PCR product as template**, resulting in **15 µL** reaction mixture for 2′ PCR.

Negative control contains only **14.25 µL** master mixture and premixed barcoded-head primers but not DNA template

15. Mix gently by tapping the tubes. Quick spin the tubes.

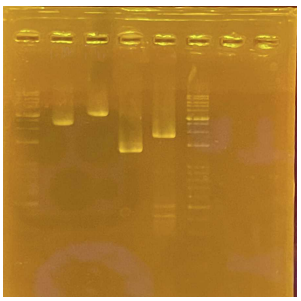
16. Carry out 2′ PCR using the following condition:

2′ PCR condition for **barcoded-head primers**

Step	Temp	Sec	Cycle
Initial denaturation	**98 °C**	30	
Denaturation	**98 °C**	15	**12 cycles**
Annealing	**64-68 °C varied (a)**	15	
Extension	**72 °C**	20 (b)	
Final extension	**72 °C**	210	
Preservation	Preservation	**4 °C**	∞

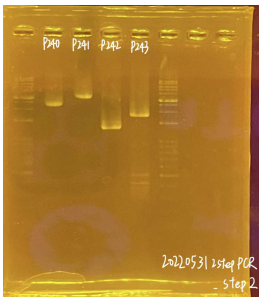
a. Annealing varied, 65 C is working based on test on 220531; Refer 2′ PCR primers for annealing temperature
b. 1kb ~ 1min extension; enough time allow full extension of sequence

17. Carry out **electrophoresis** for inspection of DNA products



Gel before markdown

18. Markdown wells and upload the pictures to the Lab Google drive



Marked gel picture go to the Lab Google drive